ECDLP lead discovery, 2026-10-24: prime-field index calculus first

Session report for RQ-ECDLP-361f3a (propose-ideas, lead-discovery round). Every statement below is labelled proposed, derived, measured (by whom) or independently verified. Nothing in this report was approved, dispatched or run as an experiment by this session (hard rule 3).

a. Orientation

- **Atlas brief (live tools, run from the CAR root at the commits below):** `ECDLP lead atlas (knowledge/leads/ecdlp/, reviewed 2026-10-24): 147 leads, 72 closed with a named obstruction, 51 open with a decided next step, 16 blocked, 8 unexamined. 2049/2078 ECDLP ledger records mapped to a lead. Changed since the review: crypto-autoresearcher: 146 paths; crypto: 16 paths; cryptanalysis: 32 paths; ecc-challenges: 3 paths.`
- **Delta status:** `--delta` lists 201 watched paths (54 decisions DEC-20261018..24, 19 CZLIFT/ICPERF/PEXT/SEMAEV hypotheses and evidence, 29 ideas IDEA-20261018..24, 12 findings/open problems, 19 problem records in CAR; 16 crypto paths; 32 cryptanalysis paths; 3 ecc-challenges paths). They were folded as **proposed row diffs** (section b and `FOLD.md`), not applied, and the review stamp was left alone (hard rule 3; AUD passes ask for diffs).
- **Commits worked at:** crypto-autoresearcher `ab0609ebee` (origin/main at 17:13 PDT; branch `lead-discovery/20261024-mythos-ecdlp`); crypto `5cb553810e` (origin/main, detached worktree); cryptanalysis `c5aa7b39b` (origin/main, detached worktree); ecc-challenges `df8cd52` at the first fetch, `86f8a57` by the end of the session; crypto-lean: **not checked out** on this host (no Lean imports were taken from it).
- **Research question:** RQ-ECDLP-361f3a (new; minted with `tools/allocate_id.py`, `--check` OK).
- **Tools:** live (`lead_atlas.py --brief/--delta/--coverage/--check --strict-paths/--render/--match`, `build_frontier_map.py --out/--match/--check`, `obstruction_registry.py --unexamined/--debt`, `build_ledger_index.py`, `validate_ledger.py`, `allocate_id.py`), not a snapshot.
- **Baseline read (crypto, origin/main, read by a delegated reader and spot-checked):** Table B prime ladder fp10..fp24 at S_IC/S_rho 5.47x (log2 r 11.9) to 13.5x (23.4), validated 13/13, no sealed bound on any Table B row; sealed prime toy IC bounds `ECBND1h93b7a9d5c187` (ic.pipeline mitm:negation_folded, alpha 0.8407 [0.6459, 0.9528]) and `ECBND1h9879c0aa348f` (subtract, alpha 1.0054 [0.8805, 1.0769]) against `rho.negation ECBND1h9896c8176a62` (alpha 0.4545 [0.3876, 0.5927]); "Prime curves have no algebraic oracle in ecbench"; Koblitz Table A at rule v8: rho 4.67-18.07x faster cold at n = 41..61, no sealed Koblitz IC bound (all six inadmissible); no end-to-end IC row on the m = 83 gate curve.

b. Audit table

(filled from the audit workflow; see `AUDIT.md` for the full per-lead record)

c. Graveyard

(filled from the scouting workflow; see `GRAVEYARD.md`)

d. Survivors, ranked

(filled from the scouting workflow)

e. Obstruction matrix and raw `--match` output

For IDEA-20261024-a2b05d (which advances a closed lead rather than opening one), the raw gate output is in `gates/gates-arity-threshold.txt`; the top hits were `LEAD-ICE-generic-oracle-arity-cost-floor` (24, 14, 24 on the three phrasings), `LEAD-ICP-generic-table-decomposition` (20), `LEAD-ICE-rootfind-pairtable-oracle` (20), `LEAD-ICP-coppersmith-smallroot-semaev` (17); frontier hits `KR-IC-5931ee` (19), `KR-RHO-13bf67` (15); ledger hits IDEA-20260803-fa9839, IDEA-20260808-2ef5c8, IDEA-20260829-b5a2c8, IDEA-20260830-3c0861, IDEA-20260830-4ab603 (the two-term model lineage). Verdict: same mechanism as the first hit; the idea upgrades its closure (AUD-3) and does not file a new lead.

f. Lean status

theorem (file)	file or attempt dir	role	status	open bridges
`CryptoResearch.IC.ArityOracleThreshold` (19 theorems: `costAll_ge_small`, `costAll_ge_large`, `costAll_ge_one`, attainment x2, `small_threshold`, `large_threshold`, `free_oracle`, `linear_oracle`, `quadratic_oracle`, `mitm_never_sub_generic`, `exhaustive_ge_one`, `attemptsAll_le_attemptsOne`, `costAll_le_costOne`, `costOne_ge_small`, `costOne_small_attained`, `costOne_ge_enum`, `costOne_enum_attained`, `main_hyps_satisfiable`; controls `needs_H1..H4`)	`formal/CryptoResearch/IC/ArityOracleThreshold.lean`; target `formal/targets/ic-arity-oracle-threshold-20261024.yaml` (TASK-20261024-9fbb0f)	cost-model algebra (rule-out within the model H1-H4)	compiles under plain `lean` v4.35.0-rc1 / Mathlib 7fa22735: 0 errors, 0 `sorry`, every `#print axioms` = propext, Classical.choice, Quot.sound; **repository gate not run to completion** (problems/formal-lean-external-volume-eintr.md); `needs_semantic_review = true`	H1-H4 are hypotheses; no bridge to curve points, counts or finite sizes; the one-decomposition semantics for 1 < k is not interpolated (only k <= 1 and k >= 1 bounds)
`CryptoResearch.IC.MemoryCouplingArityTwo` (7 theorems + control)	`formal/CryptoResearch/IC/MemoryCouplingArityTwo.lean`; target `formal/targets/ic-memory-coupling-arity-two-20261024.yaml` (TASK-20261024-2be6ac)	cost-model algebra (IDEA-20260830-4ab603's correction)	same: compiles under plain `lean`, 0 errors, 0 `sorry`, standard axioms; gate not run	H5 (B^2 table) is a hypothesis; unstructured bases only

`CryptoResearch.IC.ExtensionFieldEmbedding` (AUD-3 upgrade for `LEAD-ICP-extension-field-embedding`, kind cost_model) | `formal/CryptoResearch/IC/ExtensionFieldEmbedding.lean`; target `formal/targets/ic-extension-field-embedding-20261024.yaml` (TASK-20261024-551bd9) | cost-model algebra | compiles under plain `lean` (see `checks/`); gate not run | H6 (Gaudry's heuristic cost p^(2-2/n+o(1)) on E(F_{p^n}), KR-IC-46d822) is a hypothesis |

Derivation for the third target (source of its claim). Embedding the prime-field problem on E(F_p) (order about p) into E(F_{p^n}) and running Gaudry's decomposition with the subfield base {x in F_p} costs, under H6, p^(2-2/n+o(1)) for every fixed n >= 2 (KR-IC-46d822: q^(2-2/n) with q = p). The reference is rho on the original group at p^(1/2). Claim: for every real n >= 2, 1/2 < 2 - 2/n;

equivalently there is no n >= 2 with 2 - 2/n <= 1/2, so the embedding enlarges the problem faster than the subfield base helps, at every extension degree. Control: without the hypothesis n >= 2 the statement fails at n = 1 (2 - 2 = 0), which is the trivial extension and not an embedding, so the degree hypothesis is load-bearing. Non-vacuity: n = 2 gives the exponent 1, which is the LEAD row's own "p at n = 2". Proved in the file as `gaudry_exponent_gt_half`, `embedding_never_helps`, `needs_degree_two`, `main_hyps_satisfiable`.

Fidelity items (sheriff B3 a-e), all files: (a) statement paired with the informal claim clause by clause in the file header and in `IDEA-20261024-a2b05d.notes.md` sections 3 and 8, forall/exists order written out in the IDEA's `quantifier_order`; (b) `main_hyps_satisfiable` in each file; (c) no new computable definitions (the definitions are real-valued `max` expressions; no `#eval` differential test applies); (d) no data enters; (e) scope: real m >= 2, k >= 0, beta > 0, exponents only; never cited at any finite size.

g. Experiment specs awaiting approval (approved_by: null)

1. `specs/EXP-SPEC-oracle-order-census.md` - prime-field decomposition-oracle order census (fitted k per implemented oracle against the threshold line; positive controls with k = 1 by construction; matched rho.negation; A/A).
2. (survivors' first steps, filled from the scouting workflow)

h. Blockers (problems/ records)

- `problems/formal-lean-external-volume-eintr.md` (new): Lake/`lean` reads on the external volume fail intermittently with EINTR; the Lean gate could not be run to completion; the theorems compile under plain `lean` with retries.
- `problems/README.md` is held by Conductor task T-5065 (another session), so the index line for the new problem record is **not** added here; proposed line: `- [Lean/Lake file reads on the external APFS volume fail intermittently with EINTR, blocking the formal gate](formal-lean-external-volume-eintr.md) - open`.
- `gh auth status`: the keyring token for github.com is invalid on this host and `api.github.com` is outside the sandbox allowlist; the PR for this branch is opened only if `gh` works at delivery time, otherwise the pushed branch is reported (propose-ideas skill, "Branch and PR hygiene").
- Pre-existing: `tools/validate_ledger.py` reports `FAIL: 104 new validation error(s)` on origin/main records (IDEA-20261003-, *IDEA-20261016-*, experiments/EXP-AES-14352a, EXP-ICPERF-*, ...), none naming this session's records (tracked by `problems/full-ledger-validation-errors.md`); `lead_atlas.py --check --strict-paths` reports 2 errors on `LEAD-RHO-pool-class-quotient` (ecc-challenges evidence paths that exist only on branch `claude/ecdlp-leads-analysis-l7f87y`, commit 1f49962; proposed diff: add `ref: claude/ecdlp-leads-analysis-l7f87y` to those two evidence entries, or merge that branch).

i. Verdict lines

(filled at the end)

Checks

check	result
`python3 tools/validate_ledger.py`	FAIL, 104 pre-existing errors on origin/main; 0 errors name RQ-ECDLP-361f3a, IDEA-20261024-a2b05d or the new targets (full log in `checks/validate_ledger_full.log`)
`python3 tools/lead_atlas.py --check --strict-paths`	FAIL, 2 pre-existing errors (LEAD-RHO-pool-class-quotient evidence[4], [5]); 0 attributable to this session; without `--strict-paths` these are warnings and the check passes
`python3 tools/build_frontier_map.py --check`	PASS: `known-results map OK: 52 rows`
`lake build / AxiomAudit.lean / verify_formal_targets.py`	NOT RUN TO COMPLETION (EINTR on this host); plain `lean` compile of both new files: PASS (logs in `checks/`)